

Rishi Raval

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EDUCATION

Adani University

Bachelor of Technology in Computer Science (AI-ML)

- CGPA: 8.69/10.0** | Coursework: DSA, OS, Theory of Computation, Comp. Networks

Ahmedabad, Gujarat

Aug 2023 – May 2027

American School of Baroda

Higher Secondary Education

- Academic Achievements: **Grade 12** - 82.6% | **Grade 10** - 94.0%

Baroda, Gujarat

Graduated May 2023

TECHNICAL SKILLS

Languages: C++, C, Python, Java, JavaScript, SQL **Web:** React, HTML/CSS
Quant & ML: Black-Scholes Modeling, NumPy, Pandas, Scikit-learn, Matplotlib
Tools/OS: Git, Linux (Fedora, Kali, Arch), VS Code, Jupyter Notebooks

PROJECTS

FastGreeks — High-Performance Quant Library | C++, Templates, Math

[\[GitHub\]](#)

- Developed a **header-only C++ library** for real-time calculation of Option Prices and Greeks (Delta, Gamma, Vega, Theta, Rho) using the Black-Scholes-Merton model.
- Implemented an **Implied Volatility (IV) solver** using the Newton-Raphson method with optimized numerical stability for edge cases like zero time-to-expiry.
- Utilized the **Abramowitz & Stegun approximation** for the Normal CDF to maximize execution speed, tailored for HFT applications.

Quantitative Trading & Backtesting Engine | Python, Pandas, Matplotlib

[\[GitHub\]](#)

- Engineered a backtesting framework to evaluate mean-reversion and momentum strategies (SMA, RSI) on historical equity data.
- Computed key risk-adjusted performance metrics including **Sharpe Ratio**, **Sortino Ratio**, and **Maximum Drawdown** to validate strategy robustness.

Pulse — Financial Intelligence & Community Platform | React, Python, Firebase

[\[GitHub\]](#)

- Engineered a **real-time data pipeline** to aggregate financial news and short-form content, creating a unified stream for tracking market sentiment.
- Built an interactive trading dashboard with **technical analysis charts** and a community research hub for users to publish quantitative insights.

Market Movement Prediction System | Scikit-learn, Pandas, NumPy

[\[GitHub\]](#)

- Engineered a predictive framework utilizing ensemble methods to forecast price direction based on 90+ technical indicators.
- Implemented a feature importance pipeline that reduced dimensionality from 90 to 16 key signals, improving model generalization.

Financial NLP & Document Analysis | Python, Deep Learning

[\[GitHub\]](#)

- Developing a Deep Learning pipeline for **Sentiment Analysis** and Named Entity Recognition (NER) on SEC filings and earnings transcripts.
- Automated the extraction of risk factors and management outlook using models tailored for financial vocabulary.

CERTIFICATIONS & ACHIEVEMENTS

JPMorgan Chase & Co. – Quantitative Research Virtual Experience | Forage

Nov 2025

- Simulated the work of a Quantitative Researcher; analyzed loan books to estimate default probability.
- Utilized dynamic programming to bucket FICO scores and visualize credit risk data.

Competitive Programming | LeetCode

- Solved **550+ problems** across Easy, Medium, and Hard difficulties.
- Contest Rating: **1444** | Focus: Dynamic Programming, Graph Theory.

Career Essentials in Generative AI | Microsoft & LinkedIn

Jul 2024

- Acquired foundational skills in Generative AI and Prompt Engineering.

INTERESTS

Quant Finance: HFT Systems, Market Microstructure **Writing:** [Recursify](#) & [Rishi Writes](#) **Hobbies:** Linux, Photography